LEAP: Lead Exposure Assistance Program

Quantitative Funding Framework: Executive Summary

Municipal Infrastructure Finance Analysis

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Contents

1 The Problem 2

2 Five-Phase Modeling Framework 2

3 Eligible Universe: 24,416 Properties 2

4 Phase 1: Deterministic Baseline 3

5 Phase 2: Monte Carlo Simulation 3

6 Phase 3: Property-Level Micro-Simulation 5

7 Phase 4: Competing Risks Repayment Model 6

8 Phase 5: Weighted Meta-Distribution and Final Answer 7

9 Final Recommendation 9

1 The Problem

Columbus faces a federal mandate to replace all lead and galvanized private-side water service lines by 2037. The Lead Exposure Assistance Program (LEAP) offers homeowners a zero-interest, deferred-repayment loan of up to $10,000. Repayment occurs as a single lump sum when the property is sold, refinanced, or equity is extracted via HELOC. If inherited and never sold, repayment is deferred indefinitely — up to 99 years.

This creates an unusual financing challenge: the city must deploy tens of millions of dollars over twelve years with no predictable return schedule. We answer four questions:

1. How much upfront capital is required in 2026?
2. What funding level is sufficient with 95% probability?
3. How does uncertainty in participation, costs, and repayment affect the answer?
4. How does a loan structure compare to a grant?

2 Five-Phase Modeling Framework

We built five sequential models, each adding a layer of realism:

Phase	Type	Key Addition
1	Deterministic	Baseline cash flow model, sensitivity grid
2	Monte Carlo	Stochastic uncertainty, three breakage scenarios
3	Micro-simulation	Property heterogeneity, non-repayment quantification
4	Competing Risks	Explicit refi/HELOC channels, Vasicek interest rates
5	Meta-Distribution	Weighted 95% CI across all breakage scenarios

Data sources: Program pilot data (82 applicants, 42 completed loans, 63 contractor quotes) · WSL insurance records (2,862 policies) · System material cross-table (288,389 service line records) · CFPB HMDA 2024 (43,276 Columbus MSA mortgage records) · Census ACS B25026 (Columbus owner tenure) · SIAM-MTI simulated property dataset

3 Eligible Universe: 24,416 Properties

Customer-side pipe material is unknown for 59,973 properties. We estimate hazardous properties among the unknowns using the conditional probability $\hat{p}(u) =$ known hazardous/known total by utility material type:

Utility Material	Hazard Rate	Est. Hazardous (unknowns)
Lead / Galvanized	33%	4,701
Non-Lead Copper	10%	2,668
Unknown	17%	2,691
Other	<0.2%	3
Total		**10,062**

$$N_{\text{total}} = 14{,}354 \text{ (confirmed)} + 10{,}062 \text{ (estimated)} = \mathbf{24{,}416}$$

4 Phase 1: Deterministic Baseline

Using fixed assumptions — 51.22% participation (42/82 pilot), $7,582 average loan cost (+3%/yr), 12-year repayment lag — the deterministic model produces:

Scenario	Required Capital	Repayments by 2037
Base case ($L = 12$ yr)	$113,516,901	$0 (0%)
Optimistic ($L = 7$ yr)	$81,563,790	$33,878,654 (30%)

The critical insight: Under a 12-year average repayment lag, *not a single dollar returns before 2038*. The program behaves like a grant during its own deployment window — the loan structure provides long-run cost recovery but no near-term liquidity relief.

Sensitivity: Participation rate dominates all other variables. Moving from 33% to 89% uptake changes required capital by 2.7×. Beyond 12 years, additional repayment lag has *zero* impact on near-term capital needs.

Loan vs. Grant Comparison

Model	Upfront Capital	Long-Run Net Cost	True Economic Cost
Grant	$113,516,901	$113,516,901	$113,516,901
Loan ($L = 12$)	$113,516,901	$0	$47,677,098
Loan ($L = 7$)	$81,563,790	$0	$27,811,641

The loan program's true economic cost — the implicit financing subsidy foregone on operating reserves at 3.5%/yr — is $47.7M, less than half the grant cost. Adding Phase 3's $13M non-repayment loss gives a total true cost of approximately $60.7M, or 53% of the grant equivalent.

5 Phase 2: Monte Carlo Simulation

Phase 2 replaces fixed assumptions with probability distributions and runs 10,000 simulations per scenario:

- **Universe size:** $\mathcal{N}(24{,}416,\ 1{,}006^2)$
- **Participation rate:** Beta(42, 40) — encodes the pilot exactly: 42 completions, 40 non-completions
- **Repayment lag:** LogNormal(mean=12, std=5 yr)
- **Replacement cost:** LogNormal fit to 63 pilot quotes (KS $p = 0.41$, good fit)

The Breakage Rate Problem

The insurance dataset (2,862 WSL policies, 41 repairs over 5 months) implies a 3.44% annual breakage rate — 14× higher than the pilot rate of 0.24%. The discrepancy has two causes: (1) insurance policyholders self-select because they suspect risk, and (2) the pilot denominator (all 24,416 eligible properties) is too large. We model three scenarios:

Scenario	Rate	Derivation	95th Pct Capital
Primary	0.24%	Pilot data; denominator too large	$133,991,878
Adjusted	**1.38%**	Insurance ÷ 2.5× bias factor	**$164,705,237**
Adverse	3.44%	Raw insurance rate; no correction	$218,243,730

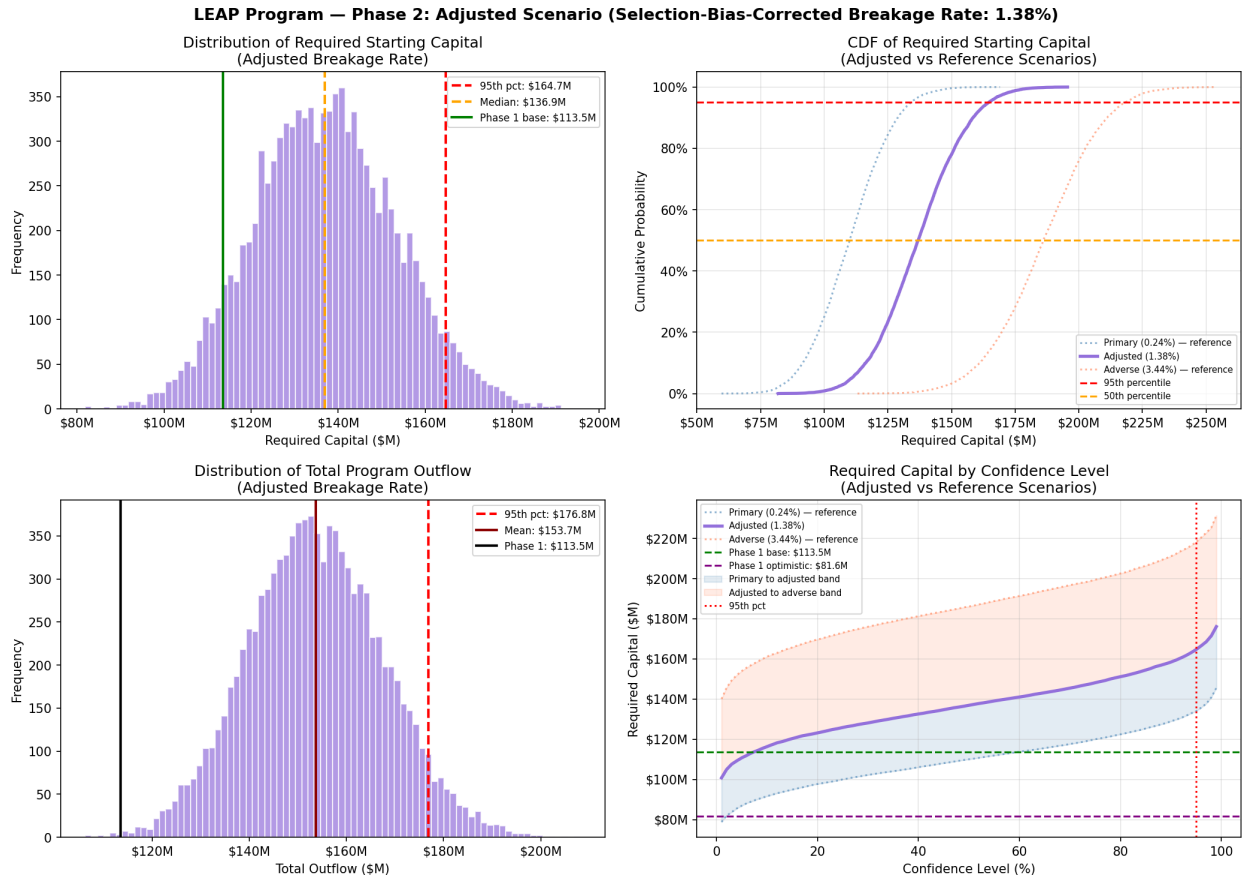

Figure 1: Phase 2 adjusted scenario (1.38% breakage, 10,000 simulations). The 95th percentile of $164.7M is our primary recommended funding level. Primary (0.24%) and adverse (3.44%) shown as dotted reference lines.

The Phase 1 estimate of $113.5M falls at the **40th percentile** of the primary distribution — insufficient 40.7% of the time. It is not a conservative number.

6 Phase 3: Property-Level Micro-Simulation

Note: Phase 3 uses a synthetically generated property dataset (SIAM-MTI). Results are structural illustrations.

Phase 3 models all 6,057 eligible properties individually using a logistic regression participation model and property-type-specific repayment lags. Its primary contribution is the quantification of **permanent non-repayment via inheritance**:

Property Type	Occupancy	Non-Repayment Rate	Mean Repayment Lag
Single family	Owner-occupied	10%	14 yr
Single family	Investor-owned	8%	10 yr
Duplex	Owner-occupied	15%	12 yr
Duplex	Investor-owned	10%	9 yr
Multi-family	Owner-occupied	20%	11 yr
Multi-family	Investor-owned	12%	8 yr
Blended		**11.57%**	

Mean permanent loss: $13.0M (12 cents per dollar loaned — never recovered)

Phase 3's 95th percentile capital of $102M appears lower than Phase 2 because it models only the voluntary participation channel. Its contribution is the $13M non-repayment finding, not a capital requirement comparison.

7 Phase 4: Competing Risks Repayment Model

Phase 4 replaces the single repayment lag with three explicit competing triggers, calibrated to real Columbus data:

- **Property sale** — Weibull hazard (k=1.17, λ=13.9 yr) fit to Census ACS B25026 Columbus tenure distribution
- **Refinancing** — 2.33%/yr at 7.44% avg rate (CFPB HMDA 2024, 43,276 Columbus records), exponential rate sensitivity
- **HELOC** — 0.75%/yr at 7.44%, modeled via Vasicek stochastic interest rate paths across three macro scenarios

The Finding

Repayment Trigger	Share of All Repayments
Property sale	38.1%
Refinancing	**38.7%**
HELOC	8.6%
Never (non-repay)	14.6%

Refinancing accounts for nearly as many repayments as property sales — a channel entirely absent from Phases 1–3. **Fed rate policy directly affects LEAP cash flows.** A rate cut cycle reduces required capital by $30–44M relative to Phase 2.

Rate Scenario	95th Pct Capital	vs. Phase 2 Primary
Low rates	\$89,779,476	−33%
Base rates	\$97,067,155	−28%
High rates	\$102,147,997	−24%
Rate sensitivity	**\$12.4M**	

Phase 4 is lower than Phase 2 because the refi and HELOC channels bring repayments *into* the program window. Phase 2 remains the conservative planning figure if equity extraction activity is uncertain.

8 Phase 5: Weighted Meta-Distribution and Final Answer

We do not know which breakage scenario is correct. Phase 5 assigns probability weights to each Phase 2 scenario and combines 100,000 samples into a single distribution — producing a true 95% CI that accounts for both within-scenario simulation uncertainty and between-scenario model uncertainty. Combining several distributions with spread means introduces fat tails, as shown in figure 2.

Scenario	Weight	Rationale
Primary (0.24%)	20%	Likely understated; denominator too large
Adjusted (1.38%)	**55%**	**Best estimate; real data with principled correction**
Adverse (3.44%)	25%	Receives weight because correction factor is itself assumed

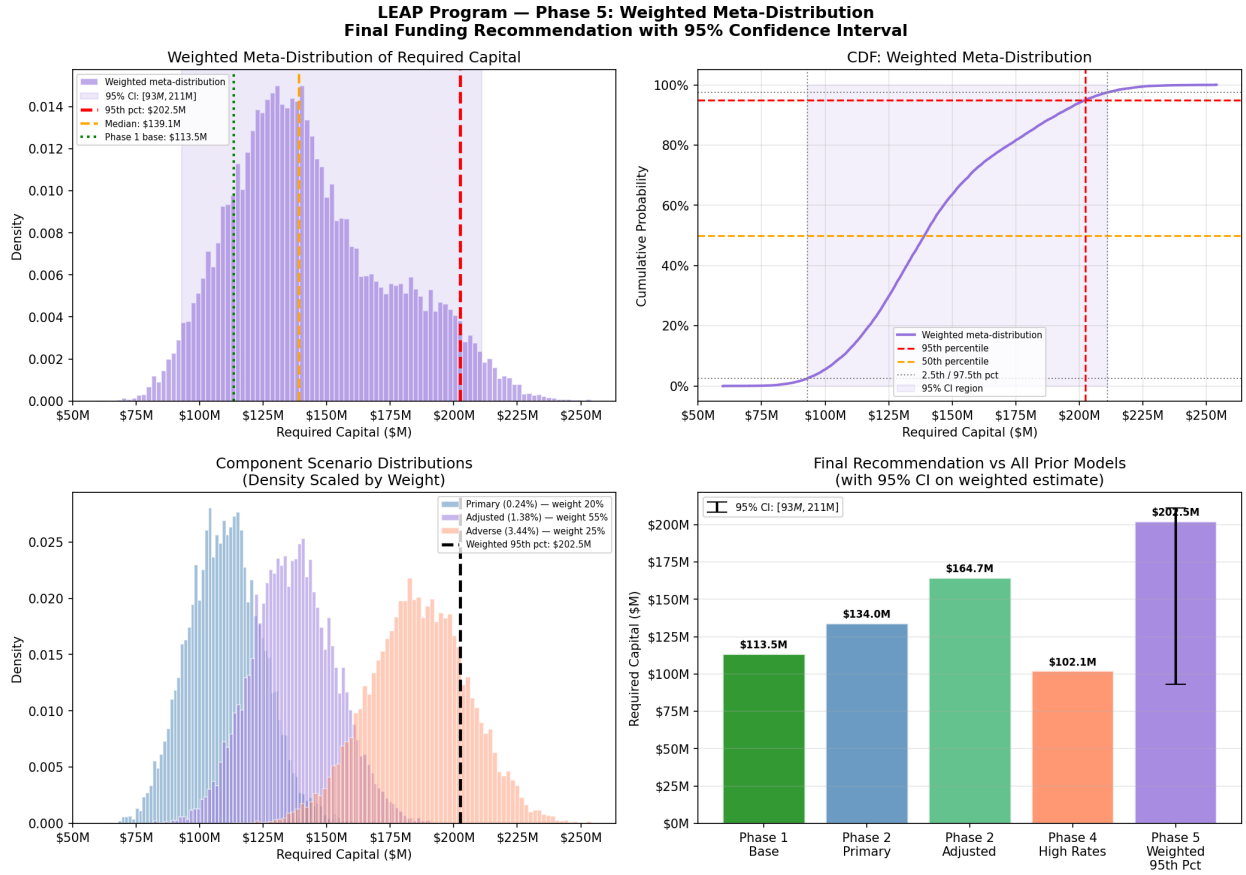

Figure 2: Phase 5 weighted meta-distribution (100,000 samples). The 95th percentile of \$202.5M and 95% CI [\$93M, \$211M] account for both simulation uncertainty and breakage rate model uncertainty. The \$37.5M gap between \$165M and \$202.5M is the quantified cost of not knowing the true pipe failure rate.

Metric	**Value**
Median	\$139,122,531
95th percentile	**\$202,459,838**
95% confidence interval	[\$93.0M, \$211.1M]

No defensible weighting scheme produces a number below \$165M. The \$37.5M gap between \$165M and \$202.5M is the cost of not knowing the true breakage rate. This could be resolved with more data on breakage rates.

9 Final Recommendation

Tier	Amount	Meaning
Conditional floor	$102.1M	Phase 4: if refi/HELOC channels active
Absolute floor	$113.5M	Phase 1 deterministic; fails 40% of the time
Lower bound	$134.0M	Phase 2 primary; breakage likely understated
Recommended	**$164.7M**	**Phase 2 adjusted; 95% confidence, bias-corrected**
Risk-averse	$202.5M	Phase 5 weighted; 95% across all scenarios
Stress test	$218.2M	Phase 2 adverse; no selection bias correction

95% Confidence Interval (Phase 5): [$93M, $211M]

Policy Actions

1. **Fund $165M** as the baseline capital reserve. Consider $202.5M if the city wants 95% confidence even accounting for model uncertainty.
2. **Structure as loans, not grants.** True economic cost of loans ($60.7M) is 53% of the grant equivalent ($113.5M).
3. **Budget a $13M non-repayment reserve.** Permanent losses from inheritance are real and quantifiable.
4. **Prioritize participation outreach.** The dominant driver of capital needs — a 33% to 89% uptake shift changes required capital by 2.7×.
5. **Monitor housing turnover and Fed rate policy** annually as leading indicators of repayment timing.
6. **Collect more breakage data.** Worth up to $37.5M in reduced capital reserve requirements.